

2026 Spring Cell Biology Homework (I)

- Your answers are due: **5 pm, April 13th, 2026.**
- Provide your answers on the answer sheet provided. Be sure to fill in your name and student ID number.
- The full score is 100.

I. Multiple choice questions (Select the best answer unless instructed otherwise.

Questions 1~38, 2 points each; questions 39~50, 1 point each.)

1. Which of the following best defines orthologous genes?
 - a. Genes duplicated within the same genome
 - b. Genes in different species derived from a common ancestor
 - c. Genes encoding identical proteins
 - d. Genes expressed in the same tissue
2. Which of the following techniques is used to detect protein–protein interactions in living cells?
 - a. Western blot
 - b. FRET
 - c. PCR
 - d. Southern blot
3. Which of the following best describes the function of the Golgi apparatus?
 - a. Protein degradation
 - b. Protein folding
 - c. Protein modification and sorting
 - d. DNA replication
4. Which cytoskeletal element is most important for maintaining cell tensile strength?
 - a. Actin filaments
 - b. Microtubules
 - c. Intermediate filaments
 - d. Centrosomes
5. Which of the following is NOT a function of the cytoskeleton?
 - a. Cell shape maintenance
 - b. Intracellular transport
 - c. DNA replication
 - d. Cell motility
6. Which of the following best describes the role of ATP in motor proteins?
 - a. Structural support
 - b. Energy source for movement
 - c. DNA binding
 - d. Signal transduction

7. Which of the following is a likely mechanism for rapid spread of antibiotic resistance?

- a. Gene duplication
- b. Horizontal gene transfer
- c. Random mutation
- d. DNA repair

8. Microtubules exhibit dynamic instability due to:

- a. ATP hydrolysis
- b. GTP hydrolysis
- c. DNA replication
- d. Actin polymerization

9. Which of the following proteins regulates actin branching?

- a. Tubulin
- b. Arp2/3 complex
- c. Lamin
- d. Dynein

10. The power stroke in myosin movement is associated with:

- a. ATP binding
- b. ATP hydrolysis
- c. Pi release
- d. ADP binding

11. A key feature of transformed cell lines is:

- a. Normal karyotype
- b. Finite lifespan
- c. Anchorage dependence
- d. Genomic instability

12. The resolution limit of a light microscope ($\sim 0.2 \mu\text{m}$) is determined by:

- a. Magnification
- b. Wavelength of light
- c. Detector sensitivity
- d. Sample thickness

13. Which structure is near the resolution limit of light microscopy?

- a. Ribosome
- b. Virus
- c. Mitochondrion
- d. DNA helix

14. Which statement is NOT part of cell theory?

- a. Cells arise from pre-existing cells
- b. Cells are fundamental units of life
- c. All cells have nuclei
- d. Organisms consist of cells

15. Which property makes yeast especially powerful for genetic screening?

- a. Large genome
- b. Rapid generation time
- c. Multicellularity
- d. High mutation rate
- e. Lack of nucleus

16. A major advantage of using *Drosophila* as a model organism is:

- a. Diploid genome simplicity
- b. Transparent body
- c. Well-characterized developmental genetics
- d. Lack of nervous system
- e. Long lifespan

17. Which model organism is most suitable for studying apoptosis pathways genetically?

- a. *E. coli*
- b. Yeast
- c. *C. elegans*
- d. Mouse
- e. Frog

18. Fluorescence Activated Cell Sorting (FACS) analysis separates cells primarily based on:

- a. Size only
- b. Density
- c. Fluorescence intensity
- d. Charge
- e. Shape

19. Why are myeloma cells used in hybridoma formation?

- a. High antibody diversity
- b. High mutation rate
- c. High antigen specificity
- d. Fast migration
- e. Immortality

20. Actin polymerization is critical for:

- a. Cell migration
- b. DNA replication
- c. Protein translation
- d. Lipid synthesis
- e. Nuclear division

21. Microtubule nucleation occurs at:

- a. Golgi
- b. Nucleus
- c. Lysosome
- d. Centrosome
- e. Ribosome

22. Which protein caps the minus end of actin filaments?

- a. CapZ
- b. Nebulin
- c. Actin
- d. Titin
- e. Tropomodulin

23. Which of the following best explains how titin contributes to muscle elasticity?

- a. It binds ATP
- b. It forms rigid filaments
- c. It unfolds under tension and recoils
- d. It caps actin filaments
- e. It degrades proteins

24. Why do phospholipids spontaneously form bilayers in water?

- a. Hydrophobic effect
- b. Ionic interactions
- c. Hydrogen bonding
- d. Covalent bonding
- e. ATP hydrolysis

25. Which lipid carries a net negative charge at physiological pH?

- a. Phosphatidylcholine
- b. Sphingomyelin
- c. Cholesterol
- d. Phosphatidylserine
- e. Glycolipid

26. Which structure forms when amphipathic lipids close into a sphere?

- a. Micelle
- b. Vesicle
- c. Fiber
- d. Lamella
- e. Crystal

27. Membrane asymmetry is important because it:

- a. Prevents diffusion
- b. Enables functional specialization
- c. Stops protein folding
- d. Blocks signaling
- e. Prevents lipid synthesis

28. Glycosylation occurs mainly in:

- a. Golgi
- b. Nucleus
- c. Cytosol
- d. Mitochondria
- e. Lysosome

29. Protein-free lipid bilayers are impermeable to:

- a. Small hydrophobic molecules
- b. CO₂
- c. Ions
- d. O₂
- e. Steroids

30. Transporters differ from channels because they:

- a. Form pores
- b. Are faster
- c. Are always open
- d. Transport only ions
- e. Undergo conformational changes

31. Electrochemical gradient consists of:

- a. Temperature + pressure
- b. Voltage + concentration
- c. pH + ATP
- d. Lipids + proteins
- e. DNA + ions

32. Ion channels are selective due to:

- a. Lipid composition
- b. Selectivity filter
- c. ATP binding
- d. DNA interaction
- e. Cytoskeleton

33. Detergents solubilize membrane proteins because they are:

- a. Charged
- b. Hydrophobic
- c. Amphipathic
- d. Enzymatic
- e. Polar

34. Short tandem repeats (STRs) markers are widely used in forensic science because they:

- a. Encode proteins
- b. Are highly conserved
- c. Show high variability between individuals
- d. Are located only in mitochondria
- e. Are identical in all humans

35. Short tandem repeat (STR) loci are typically found in:

- a. Only coding regions
- b. Only mitochondrial DNA
- c. Noncoding regions of the genome
- d. Ribosomal RNA
- e. Proteins

36. Which condition is required for Fluorescence Resonance Energy Transfer (FRET) to occur?

- a. Same emission spectra
- b. ATP presence
- c. High temperature
- d. Overlap between donor emission and acceptor excitation
- e. Protein degradation

37. Which of the following statements about membranes are correct? (*Select all that apply*)

- a. Membrane fluidity depends on lipid composition
- b. Membrane proteins can diffuse laterally
- c. Membranes are asymmetric
- d. Lipid flip-flop occurs rapidly without enzymes

38. Which of the following statements about actin dynamics are correct? (*Select all that apply*)

- a. Actin filaments have polarity
- b. ATP-actin polymerizes more efficiently than ADP-actin
- c. Actin filaments are static structures
- d. Profilin promotes actin polymerization
- e. Capping proteins regulate filament growth

39. Proteins can be separated by size on SDS-PAGE because they are uniformly negatively charged by _____.

- a. Sodium Dodecyl Sulfate
- b. beta-mercaptoethanol
- c. Both
- d. None of these

40. Can mass spectrometry be used to distinguish post-translational modifications of proteins, such as phosphorylation?

- a. Yes
- b. No

41. Is protein crystallization required to determine protein structure using X-ray diffraction?

- a. Yes
- b. No

42. Which of the following is not a restriction nuclease?

- a. HindIII
- b. EcoRI
- c. Taq

43. Which type of library should be constructed to clone the entire DNA content of the human genome?

- a. Genomic library
- b. cDNA library
- c. EST library

d. Sequencing library

44. DNA amplification by PCR is _____.

- a. Exponential
- b. Linear
- c. Sequential

45. The key feature of Sanger sequencing is the incorporation of _____ to terminate DNA chain elongation.

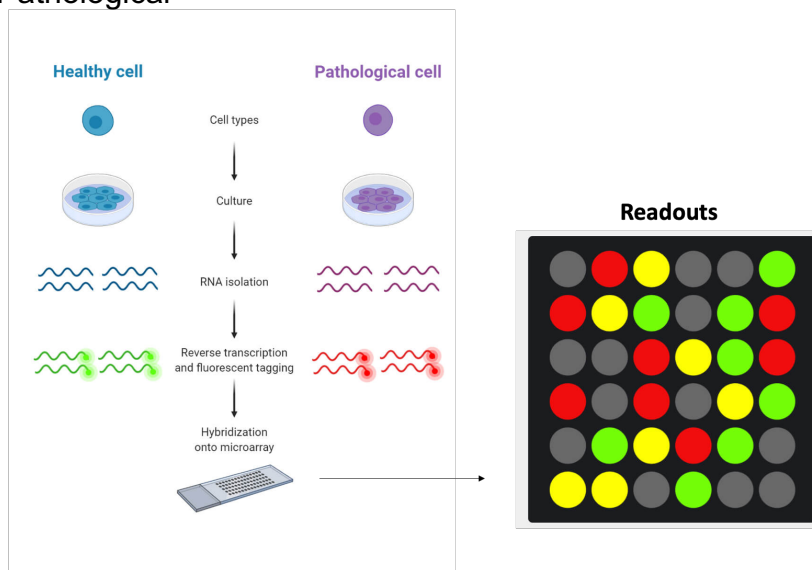
- a. Deoxynucleotide triphosphates
- b. Dideoxynucleotide triphosphates
- c. Nucleoside triphosphates

46. The major difference between DNA sequence polymorphisms and mutations is that polymorphisms are _____ in the population.

- a. Commonly present
- b. Random
- c. Pathological

47. In the microarray shown below, red dots represent genes that are more highly expressed in _____ cells.

- a. Healthy
- b. Pathological



48. To separate small particles such as ribosomes from a cellular homogenate, you would require a _____ centrifugation speed to generate _____ gravitational force.

- a. Lower, higher
- b. Higher, higher
- c. Higher, lower
- d. Lower, lower

49. If you aim to purify a protein based primarily on its size, which chromatographic method should you choose first?

- a. Ion-exchange chromatography
- b. Gel-filtration chromatography

c. Affinity chromatography

50. Formins primarily promote actin filament assembly by:

- a. Severing actin filaments
- b. Nucleating branched actin networks
- c. Depolymerizing actin
- d. Capping the minus end
- e. Nucleating and elongating linear actin filaments

II. Short-answer questions

1. A researcher uses FRAP to compare two GFP-tagged plasma membrane proteins. After photobleaching, Protein A shows rapid and nearly complete fluorescence recovery, whereas Protein B shows slow and incomplete recovery. What do these results indicate about the mobility and membrane association of the two proteins? Explain your reasoning. (6 points)

2. Researchers measure membrane fluidity using a fluorescence probe in two cell types at both 37°C and 10°C. At 37°C, both Cell A and Cell B show similar membrane fluidity. However, at 10°C, Cell A maintains relatively high membrane fluidity, while Cell B shows a dramatic decrease in fluidity. Lipid analysis reveals that Cell A contains a higher proportion of unsaturated fatty acids compared to Cell B.

(1) Based on these data, which type of fatty acid helps maintain membrane fluidity at low temperature? (2 points)

(2) Explain why this difference becomes apparent specifically at low temperature. (4 points)